GG-I-526b
October 11, 1965
SUPERSEDING
Fed. Spec. GG-I-526a
April 8, 1949

FEDERAL SPECIFICATION

# INSTRUMENTS, DENTAL AND SURGICAL; GENERAL SPECIFICATIONS

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements and tests for dental and surgical instruments, and is intended to be used in conjunction with the detail specification, purchase description, or procurement document, covering such additional requirements as are applicable to each instrument.

1.2 Classification.

1.2.1 *Classes.* Instruments covered by this specification shall be of the following classes, as specified (see 6.1):

1—Carbon steel.
2—High carbon steel.
3—Austenitic stainless steel.
4—Martensitic stainless steel.
5—Precipitation hardening stainless steel.
6—Ferritic stainless steel.
7—Cobalt-chromium-tungsten alloy.
8—Nonferrous metals
9—Unclassified.

1.2.2 *Types.* Materials for instruments covered by this specification shall be of the types specified in tables I through IX (see 6.1).

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standard, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

*Federal Specifications:*

QQ-C-320—Chromium Plating (Electrodeposited).
QQ-N-290—Nickel Plating (Electrodeposited).

*Federal Standard:*

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 6515



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*American Pharmaceutical Association Publication*:

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

*American Chemical Society Publication*:

Reagent Chemicals.

(Application for copies should be addressed to Applied Publications, American Chemical Society, Washington, D. C.)

## 3. REQUIREMENTS

3.1 Material. The material shall conform to tables I through IX for the class and type specified, and shall be so treated and otherwise processed as to result in finished instruments in full compliance with the requirements specified hereinafter and in the detail specification (see note below). Instruments shall not be plated unless so required herein or in the detail specification. Material for class 9 shall be as specified in the detail specification (see 6.1).

Note: All reference to "detail specification" are intended to include formal or interim specifications, purchase descriptions, procurement documents, or any other document utilized by the procuring activity.

3.2 Hardness. The operating parts of the finished instrument such as cutting edges, jaws, and surfaces subject to wear shall be of the Rockwell hardness specified in the detail specification (see 6.1). Mating surfaces on the same instrument such as opposite blades and jaws, shall not vary in hardness by more than 5 points (Rockwell). Other parts of cutting instruments need not be fully hardened. The finished instrument shall comply with all corrosion resistance and workmanship requirements (see 3.7 and 3.9), regardless of its hardness.

3.2.1 *Stress relieving and annealing.* Instruments shall be properly annealed, or stress relieved, and otherwise treated, as applicable for the material involved, to obtain maximum corrosion resistance.

3.3 Distortion. The shanks, jaws, or any other part of the ratchet type instruments shall not break or take a permanent change in shape when tested as specified in 4.4.3.

3.4 Style, design, and dimensions. Instruments shall conform to style, design, and dimensions specified in the detail specification (see 6.1). Tolerances indicated in drawings and figures are intended to delineate extreme dimensions, and are not intended to permit variations within a single instrument which will sacrifice proper uniformity, symmetry, balance, and first quality workmanship.

3.5 Construction.

3.5.1 *Finger rings.* Finger rings shall be of proper size and shape for maximum utility and comfort. Inside of finger rings shall be well rounded. Finger rings shall be free of sharp edges, rough areas, grind marks, cracks, overlaps, burrs, or other defects.

3.5.2 *Jaw serrations.* Jaw serrations, shall be well cut and defined; cross serrations and longitudinal serrations shall mesh properly when jaws are closed. The ends of the serrations shall be well chamfered. Jaws shall not contain burrs or sharp edges.

3.5.3 *Handle serrations.* Handle serrations shall be properly formed and shall provide a good gripping surface when instrument is used for purpose intended.

3.5.3.1 *Fillet serrations.* Serrations shall be of sufficient depth to provide a satisfactory gripping surface, suitable for purpose intended.

3.5.4 *Teeth.* Unless otherwise specified, teeth shall be sharp, of proper size and shape, free from rough edges or burrs, and shall mesh with sufficient accuracy to insure proper performance for the use intended (see 6.1).

3.5.5 *Ratchet and ratchet catches.* Ratchet and ratchet catches shall be properly aligned and undercut for safe locking. Ratchets shall be of such design as to insure easy and positive engagement and proper



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disengagement. Ratchets and ratchet catches shall be free of burrs and sharp edges.

3.5.6 *Locks.* Forceps and similar instruments shall be of the screw type, box lock type, or lap joint type, as specified (see 6.1). All types of locks shall be accurately fitted, without stiffness, and without crevices, burrs, or sharp edges at any place in their construction. Pins and screws shall be permanently secured. When specified (see 6.1), shanks of instruments tested in accordance with 4.4.6, shall fall freely.

3.5.6.1 *Screws.* The screws of screw lock forceps, scissors, and other instruments shall be concentric shouldered type, countersunk, flush with, or slightly below, the surface, or rounded, smooth, and flush at the periphery, but shall not be riveted. Screw shall retain position after setting without binding or loosening during use.

3.5.7. *Cutting edges.* Cutting edges shall be properly sharpened to the required type of edge and shall be free from burrs or feathered edges. The cutting instruments shall function correctly for the particular work for which they are intended. Blades of cutting instruments (inside blade only for scissors) shall have a crocus finish as specified in 3.6.4, or a highly polished finish. Scissors, when tested as specified in 4.4.2, shall cut a continuous length of the test material of at least the length specified in the detail specification (see 6.1), This cutting requirement shall apply for the entire length of the cut specified therein including the tips of the scissors. All cuts made with each scissors tested shall meet the requirements of the detail specification.

3.6 Finish. Finish on all edges and surfaces shall be uniform, and free of burrs, sharp edges (except where required), crevices, grind marks, rough areas, cracks, and overlaps. In addition, finish shall be as hereinafter specified (see 6.1).

3.6.1 *Plated instruments.* Instruments for which nickel and chromium plating is required shall be plated as specified in table IX, in accordance with QQ-N-290 and QQ-C-320. Instruments for which silver plating is required shall be plated as specified in 3 6.1.1. The final plating shall be uniform in thickness and appearance; shall be free from blisters, visible porosity, or other defects. Instruments with cutting edges shall have the cutting edges ground to remove all plating and shall be properly honed. The base metal to be plated shall be free from flaws or defects which will be detrimental to the final finish, and shall be subjected to such polishing, cleaning, pickling, or dipping, and plating procedures as are necessary, to yield deposits with the desired appearance and quality. Soldered connections on multipiece instruments shall be plated over.

3.6.1.1 *Silver plating.* Instruments that require silver plating shall be plated with electrodeposited coatings of silver. The base metal shall be given a preliminary coating which will enable the final silver deposit to adhere to and completely cover the coated basic metal. The preliminary coating shall be one or two of the following: silver from silver strike solutions, nickel plate, or copper plate. The silver coatings of the finished instruments shall be not less than 0.0010 inch in thickness.

3.6.2 *Satin.* Instruments specified to have satin finish shall be ground to remove all surface imperfections, followed by a finish equivalent to that produced with No. 200 emery and a soft wheel. Finish shall result in smooth uniform surface with reduced reflection.

3.6.3 *Mirror.* Instruments specified to have mirror finish shall be properly ground to remove all surface imperfections and polished to remove grind marks, resulting in a mirror polish.

3.6.4 *Crocus.* Blades of cutting instruments required to have crocus finish shall have a smooth finish, with polishing marks in one direction, perpendicular to cutting edge. Finish shall be equivalent to that produced with grit No. 240.



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3.6.5 *Passivation.* As a final finishing operation, instruments, fabricated from class 4 or 5 material shall be properly passivated.

3.7 Corrosion resistance.

3.7.1 *Resistance to copper sulfate test.* There shall be no plating of copper on the instruments, when tested as specified in 4.4.4.1. Copper plating at the periphery of the drops of the copper sulfate solution, or at soldered or brazed junctions, or dulling of polished surfaces caused by the copper sulfate solution, shall be disregarded.

3.7.2 *Resistance to salt-spray test.* When specified (see 6.1), instruments shall be submitted to the salt-spray test. There shall be no corrosion spots visible to the naked eye on any significant surface of the instruments subjected to the salt-spray test specified in 4.4.4.2. The significant surface shall be as specified in the detail specification (see 6.1).

3.7.3 *Resistance to boil test.* On instruments subjected to the boil test specified in 4.4.4.3, all exposed surfaces required to be smooth shall show no signs of corrosion. A slight corrosion in serrations, teeth, locks, ratchets, crevices, etc., shall not be cause for rejection.

3.8 Identification marking. Instruments made in U.S.A. shall be marked with the name or registered trademark of the contractor, "stainless" (when applicable), and other information as may be specified in the detail specification. Instruments of foreign (other than U.S.) manufacture shall be marked with the name of the country of origin, together with the name or registered trademark of the contractor, and other information as may be specified in the detail specification (see 6.1).

3.9 Workmanship. Workmanship shall be first class throughout. Instruments shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use. Instruments shall be finished so as not to retain foreign matter, blood, pus, body fluids, etc. (i.e., absence of crevices, cracks, pits, etc.).

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling for examination and test shall be as specified in the detail specification (see 6.1).

4.3 Examination. The instruments shall be examined to determine compliance with all requirements contained in this specification and in detail specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 *Hardness test.* Hardness shall be determined in accordance with methods 243



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(Rockwell) or 244 (Diamond Pyramid) of Fed. Test Method Std. No. 151, or by other methods as may be specified in the detail specification, or by the procuring activity (see 6.1).

4.4.2 *Scissors performance test.* The test material shall be as specified in the detail specification (see 6.1). Scissors to be tested shall be cleaned prior to test performance. Three separate, consecutive tests shall be performed for each scissors. Each test shall consist of making one cut through the test material. Each cut shall be made using the scissors in the normal manner (i.e., with the examiner's fingers in the finger rings of the instrument) except that no lateral pressure shall be exerted.

4.4.3 *Distortion test.* Ratchet type forceps shall have jaws clamped in tightest ratchet position, for 24 hours on a piece of basswood 1/8 inch thick, or a leather strap 1/8 inch thick with a tapered edge.

4.4.4 *Corrosion tests.*

4.4.4.1 *Copper sulfate test.* Instruments of the classes and types for which the copper sulfate test is indicated shall be tested by the following method: Each sample shall be scrubbed with soap and warm water, rinsed in hot water, followed by a dip in ethyl alcohol (95 percent), or National Formulary (NF) or American Chemical Society (ACS) isopropyl alcohol (isopropanol; 2-propanol), and dried. Instruments, whenever possible, shall be completely immersed in the copper sulfate solution at a temperature of 65°±2° F. Instruments too large for complete immersion shall have partial immersion or test by drops of the solution. Copper sulfate solution shall be allowed to remain in contact for 6 minutes and shall then be wiped off with a cloth saturated with fresh water. The copper sulfate solution shall be made up as follows:

| | |
|---|---|
| Copper sulfate ($CuSO_4 \cdot 5H_2O$) ..... | 4.0 grams |
| Sulphuric acid ($H_2SO_4$, sp. gr. 1.84) .. | 10.0 grams |
| Water ($H_2O$) ..... | 90.0 ml. |

4.4.4.2 *Salt-spray test.* Instruments of the classes and compositions for which the salt-spray test is indicated shall be tested in accordance with method 811 of Fed. Test Method Std. No 151. Length of test shall be 24 hours.

4.4.4.3 *Boil test.* Instruments of the classes and types for which the boil test is indicated shall be tested by the following method: Before testing, instruments shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the instruments shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for an additional 24 hours.

4.4.5 *Composition analysis.* Specimens of materials used to fabricate the instrument, or taken from the sample, shall be subjected to analysis in accordance with methods 111 or 112 of Fed. Test Method Std. No. 151.

4.4.6 *Lock stiffness test.* Instrument shall be held in a verticle plane. The top half shall be raised to half open and then released.

## 5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in the detail specification (see 6.1)

## 6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Number and date of this specification.
(*b*) Classes required (see 1.2.1).
(*c*) Types required (see 1.2.2).
(*d*) Composition (see 3.1).
(*e*) Hardness (see 3.2).
(*f*) Style, design, and dimensions required (see 3.4).
(*g*) Teeth (see 3.5.4).
(*h*) Type of lock required and stiffness test (see 3.5.6).
(*i*) Cutting edge (see 3.5.7).



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(j) Finish required (see 3.6).

(k) Salt-spray test and significant surface (3.7.2, tables I, II & III).

(l) Identification marking required (see 3.8).

(m) Sampling required (see 4.2).

(n) Hardness test method (4.4.1).

(o) Test material for scissors performance test (see 4.4.2).

(p) Selection of applicable levels of packaging and packing (see 5.1).

6.2 Commercial and old designations. For information purposes only, listed below are the classes and types of carbon steel and corrosion-resisting steel specified herein and the commercial and old designations to which they are similar:

*Class* 1:

| Type | AISI No. | SAE No. | GG-I-526a, Apr. 8, 1949 |
|---|---|---|---|
| 1006 | — | 1006 | Comp. A |
| 1008 | C1008 | 1008 | Comp. A |
| 1010 | C1010 | 1010 | Comp. A |
| 1042 | C1042 | 1042 | Comp. B |
| 1043 | C1043 | 1043 | Comp. B |
| 1045 | C1045 | 1045 | Comp. B |
| 1065 | — | 1065 | Comp. C |
| 1070 | C1070 | 1070 | Comp. C |
| 1080 | C1080 | 1080 | Comp. D |
| 1085 | C1085 | 1085 | Comp. D |
| 1095 | C1095 | 1095 | Comp. E |

*Class* 3:

| Type | AISI No. | GG-I-526a April 8, 1949 |
|---|---|---|
| 301 | 301 | — |
| 302 | 302 | Comp. G(a) |
| 303 | 303 | Comp. G(b) |
| 304 | 304 | Comp. G(c) |
| 316 | 316 | — |
| 317 | 317 | — |

*Class* 4:

| Type | AISI No. | GG-I-526a April 8, 1949 |
|---|---|---|
| 403 | 403 | — |
| 410 | 410 | — |
| 416 | 416 | Comp. H |
| 420 | 420 | Comp. K |
| 440A | 440A | Comp. L(a) |
| 440B | 440B | Comp. L(b) |
| 440C | 440C | Comp. L(c) |

6.3 Catalog references. Any and all references to commercial types, styles, trade names, and catalogs are intended to be descriptive and not restrictive, and only to indicate to prospective bidders the kind of instruments desired.

6.4 Elements. The following symbols are used herein to designate elements as indicated:

Ag—Silver
C—Carbon
Cb—Columbium
Co—Cobalt
Cr—Chromium
Cu—Copper
Fe—Iron
Mn—Manganese
Mo—Molybdenum
Ni—Nickel
P—Phosphorous
Pb—Lead
S—Sulfur
Se—Selenium
Si—Silicon
Ta—Tantalum
V—Vanadium
W—Tungsten (Wolfram)
Zr—Zirconium

CUSTODIANS:
Army—MD
Navy—MS
Air Force—03

*Review activities*:
Army—MD
Navy—MS
Air Force—03

*Preparing activity*:
Defense Supply Agency—DM



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TABLE I. *Composition of class 1, carbon steels (percent)*

| Type | C | Mn | P (max.) | S (max.) | Plating* | Corrosion tests |
|---|---|---|---|---|---|---|
| 1006 | 0.08 max. | 0.25—0.45 | 0.040 | 0.050 | Nickel and chromium | Salt-spray** boil test |
| 1008 | 0.10 max. | 0.25—0.50 | 0.040 | 0.050 | " | " |
| 1010 | 0.08—0.13 | 0.30—0.60 | 0.040 | 0.050 | " | " |
| 1042 | 0.40—0.47 | 0.60—0.90 | 0.040 | 0.050 | " | " |
| 1043 | 0.40—0.47 | 0.70—1.00 | 0.040 | 0.050 | " | " |
| 1045 | 0.43—0.50 | 0.60—0.90 | 0.040 | 0.050 | " | " |
| 1065 | 0.59—0.71 | 0.60—0.90 | 0.040 | 0.050 | " | " |
| 1070 | 0.65—0.75 | 0.60—0.90 | 0.040 | 0.050 | " | " |
| 1080 | 0.75—0.88 | 0.60—0.90 | 0.040 | 0.050 | " | " |
| 1085 | 0.80—0.93 | 0.70—1.00 | 0.040 | 0.050 | " | " |
| 1095 | 0.90—1.03 | 0.30—0.50 | 0.040 | 0.050 | " | " |

* See table IX.

** When specified (see 6.1).

TABLE II. *Composition of class 2, high carbon steels (percent)*

| Type | C | Mn | Si | Cr | P Max. | S Max. | Plating* | Corrosion tests |
|---|---|---|---|---|---|---|---|---|
| W-1 | 1.10—1.40 | 0.25—0.50 | 0.050 max. | — | 0.040 | 0.050 | Nickel and chromium | Salt-spray** boil test |
| W-4 | 1.20—1.30 | 0.25—0.40 | 0.20—0.30 | 0.15—0.30 | 0.015 | 0.020 | — | — |

* See table IX.

** When specified (see 6.1).

TABLE III. *Composition of class 3, austenitic stainless steels (percent)*

| Type | C Max. | Cr | Ni | Mn Max. | P Max. | S | Si Max. | Other elements | Corrosion tests |
|---|---|---|---|---|---|---|---|---|---|
| 301 | 0.15 | 16.00—18.00 | 6.00— 8.00 | 2.00 | 0.045 | 0.030 max. | 1.00 | — | Boil test |
| 302 | 0.15 | 17.00—19.00 | 8.00—10.00 | 2.00 | 0.045 | 0.030 max. | 1.00 | — | Boil test |
| 303 | 0.15 | 17.00—19.00 | 8.00—10.00 | 2.00 | 0.20 | 0.15 min. | 1.00 | Mo-0.60 max. (optional) Zr-0.60 max. (optional) | Boil test |
| 304 | 0.08 | 18.00—20.00 | 8.00—12.00 | 2.00 | 0.045 | 0.030 max. | 1.00 | — | Boil test |
| 316 | 0.08 | 16.00—18.00 | 10.00—14.00 | 2.00 | 0.045 | 0.030 max. | 1.00 | Mo-2.00—3.00 | Boil test |
| 317 | 0.08 | 18.00—20.00 | 11.00—15.00 | 2.00 | 0.45 | 0.030 max. | 1.00 | Mo-3.00—4.00 | Boil test |



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TABLE IV. *Composition of class 4, martensitic stainless steels (percent)*

| Type | C | Cr | Mn Max. | P Max. | S | Si Max. | Other elements | Corrosion tests |
|---|---|---|---|---|---|---|---|---|
| 403 | 0.15 max. | 11.50—13.00 | 1.00 | 0.040 | 0.030 max. | 0.050 | | Copper sulfate boil test |
| 410 | 0.15 max. | 11.50—13.50 | 1.00 | 0.040 | 0.030 max. | 1.00 | | Copper sulfate boil test |
| 410 X | 0.20 max. | 11.50—13.50 | 1.00 | 0.040 | 0.030 max. | 1.00 | | Copper sulfate boil test |
| 416 | 0.15 max. | 12.00—14.00 | 1.25 | 0.06 | 0.15 min. | 1.00 | Mo-0.60 max. (optional) Zr-0.60 max. (optional) | Copper sulfate boil test |
| 420 | Over 0.15 | 12.00—14.00 | 1.00 | 0.040 | 0.030 max. | 1.00 | | Copper sulfate boil test |
| 420 F | 0.15 min. | 12.00—14.00 | 1.25 | — | — | 1.00 | P,S, or Se-0.07 min. Zr or Mo-0.60 max. | Copper sulfate boil test |
| | | | | | | | | Copper sulfate boil test |
| 440 A | 0.60—0.75 | 16.00—18.00 | 1.00 | 0.040 | 0.030 max. | 1.00 | Mo-0.75 max. | Copper sulfate boil test |
| 440 B | 0.75—0.95 | 16.00—18.00 | 1.00 | 0.040 | 0.030 max. | 1.00 | Mo-0.75 max. | Copper sulfate boil test |
| 440 C | 0.95—1.20 | 16.00—18.00 | 1.00 | 0.040 | 0.030 max. | 1.00 | Mo-0.75 max. | Copper sulfate boil test |

TABLE V. *Composition of class 5, precipitation hardening, stainless steel (percent)*

| Type | C | Cr | Ni | Mn | P | S | Si. | Al | Cu | Cb Ta | Corrosion tests |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 17-7PH | 0.09 max. | 16.00—18.00 | 6.50—7.75 | 1.00 max. | 0.040 max. | 0.040 max. | 1.00 max. | 0.75—1.50 | — | — | Boil test |
| 17-4PH | .07 max. | 15.50—17.50 | 3.00—5.00 | 1.00 max. | 0.040 max. | 0.030 max. | 1.00 max. | — | 3.00—5.00 | 0.15—0.45 | Boil test |

TABLE VI. *Composition of class 6, ferritic stainless steel*

| Type | C | Cr | Mn | P | S | Si | Corrosion tests |
|---|---|---|---|---|---|---|---|
| 430 | 0.12 max. | 14.00—18.00 | 1.00 max. | 0.040 max. | 0.030 max. | 1.00 max. | Copper sulfate boil test |



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TABLE VII. *Composition of class 7, cobalt-chromium-tungsten alloy*

| Type | C | Cr | Ni | W | Mn | Si | Fe | Co | Corrosion tests |
|---|---|---|---|---|---|---|---|---|---|
| 6B | 0.90—1.40 | 25.0—31.0 | 3.0 max. | 3.0—6.0 | 2.00 max. | 2.00 max. | 3.0 max. | 55.0 min. | Boil test |
| 25 | 0.15 max. | 18.5—21.5 | 9.0—11.0 | 14.0—16.0 | 1.0—2.0 | 1.0 max. | 3.0 max. | Remainder (44.0 min.) | Boil test |

TABLE VIII. *Composition of class 8, nonferrous metals percent maximum (unless shown on a range)*

| Alloy number and commercial designation | Cu | Pb | Fe | Zinc | Ni | Mn | Ag | Total other elements | Plating* | Corrosion tests |
|---|---|---|---|---|---|---|---|---|---|---|
| 110 (a) | 99.90 | — | — | — | — | — | — | — | Nickel and chromium | Salt-spray boil*** |
| | | | | | | | | | Silver | Boil |
| 122 (b) | 99.90 | — | — | — | — | — | — | P .015—0.040 | Nickel and chromium | Salt-spray boil*** |
| | | | | | | | | | Silver | Boil |
| 260 (Cartridge brass, 70 percent) | 68.5—71.5 | 0.07 | 0.05 | Rem. | — | — | — | 0.15 | Nickel and chromium | Salt-spray boil*** |
| 270 (Yellow brass, 65 percent) | 63.0—68.5 | .10 | .05 | Rem. | — | — | — | .15 | Nickel and chromium | Salt-spray boil*** |
| 360 (Free cutting brass) | 60.0—63.0 | 2.5—3.7 | .35 | Rem. | — | — | — | .50 | Nickel and chromium | Salt-spray boil*** |
| 752 (Nickel-silver 65-18) | 63.0—66.5 | .10 | .25 | Rem. | 16.5—19.5 | 0.50 | — | .50 | Nickel and chromium | Boil |
| Class A Nickel-copper | Rem. | — | 2.5 | — | 63.0—70.0** | 2.0 | — | C .3 Si .5 S .024 | Chromium | Salt-spray boil*** |
| Class B Nickel-copper | Rem. | — | 2.5 | — | 63.0—70.0** | 2.0 | — | C .3 Si .5 S .025—.060 | Chromium | Salt-spray boil*** |
| Sterling silver | Rem. | — | — | — | — | — | 92.5 | — | — | — |

* See table IX.
** Cobalt counting as nickel.
*** When specified (see 6.1).

(a) Electrolytic tough pitch copper.
(b) Phosphorus deoxidized copper, high residual phosphorus.

TABLE IX. *Plating*

| Class | Intermediate coating | | Final coating | |
|---|---|---|---|---|
| | Material | QQ-N-290 specification designation | Material | QQ-C-320 specification designation |
| 1 and 2 | Copper and nickel | Class 1, type III (KS) | Chromium | Class 1, type I |
| 8 | Nickel | Class 1, type VI (KC) | Chromium | Class 1, type I |

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